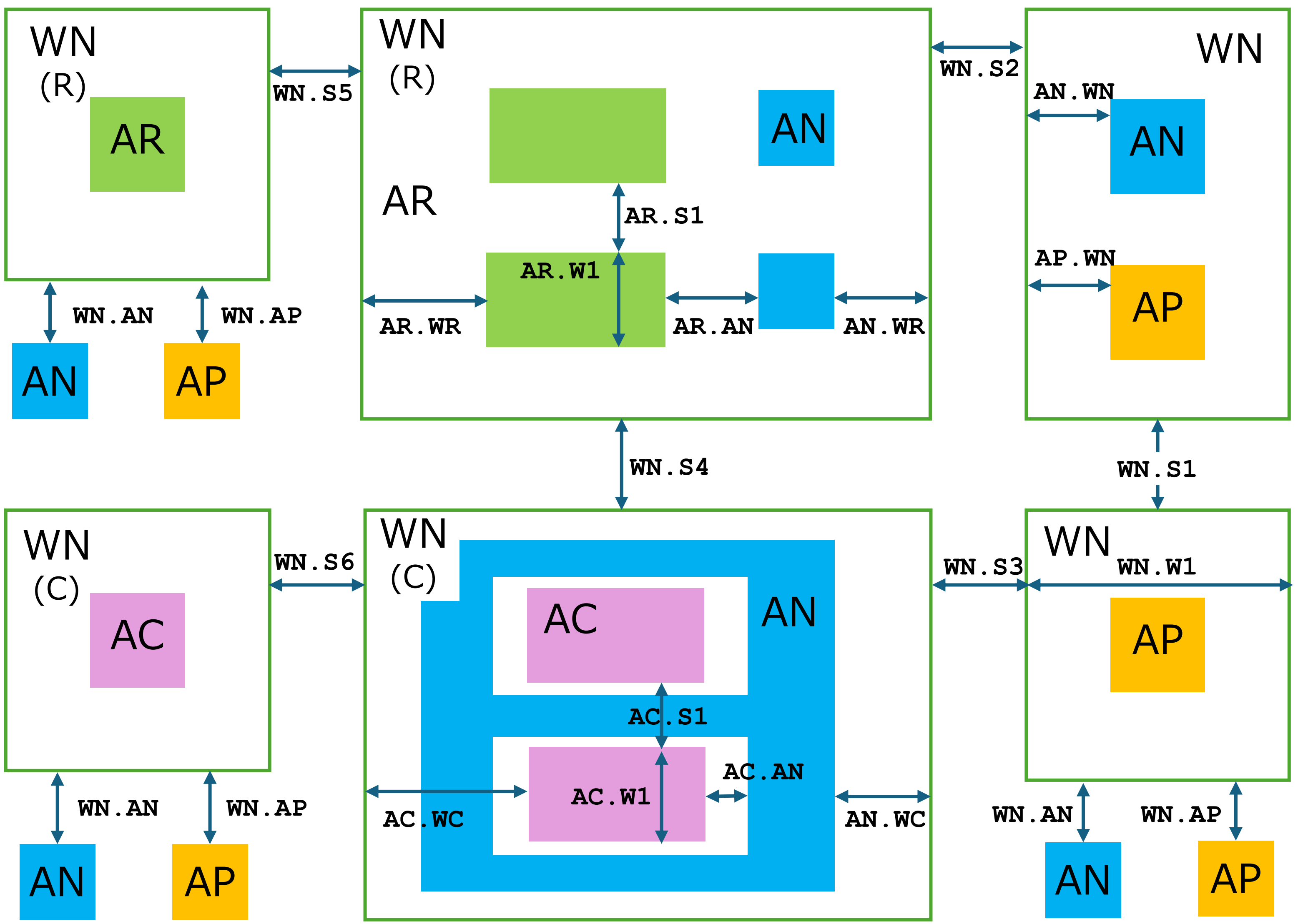
Design Rules

for Drawing Layers

Drawing Layer Table

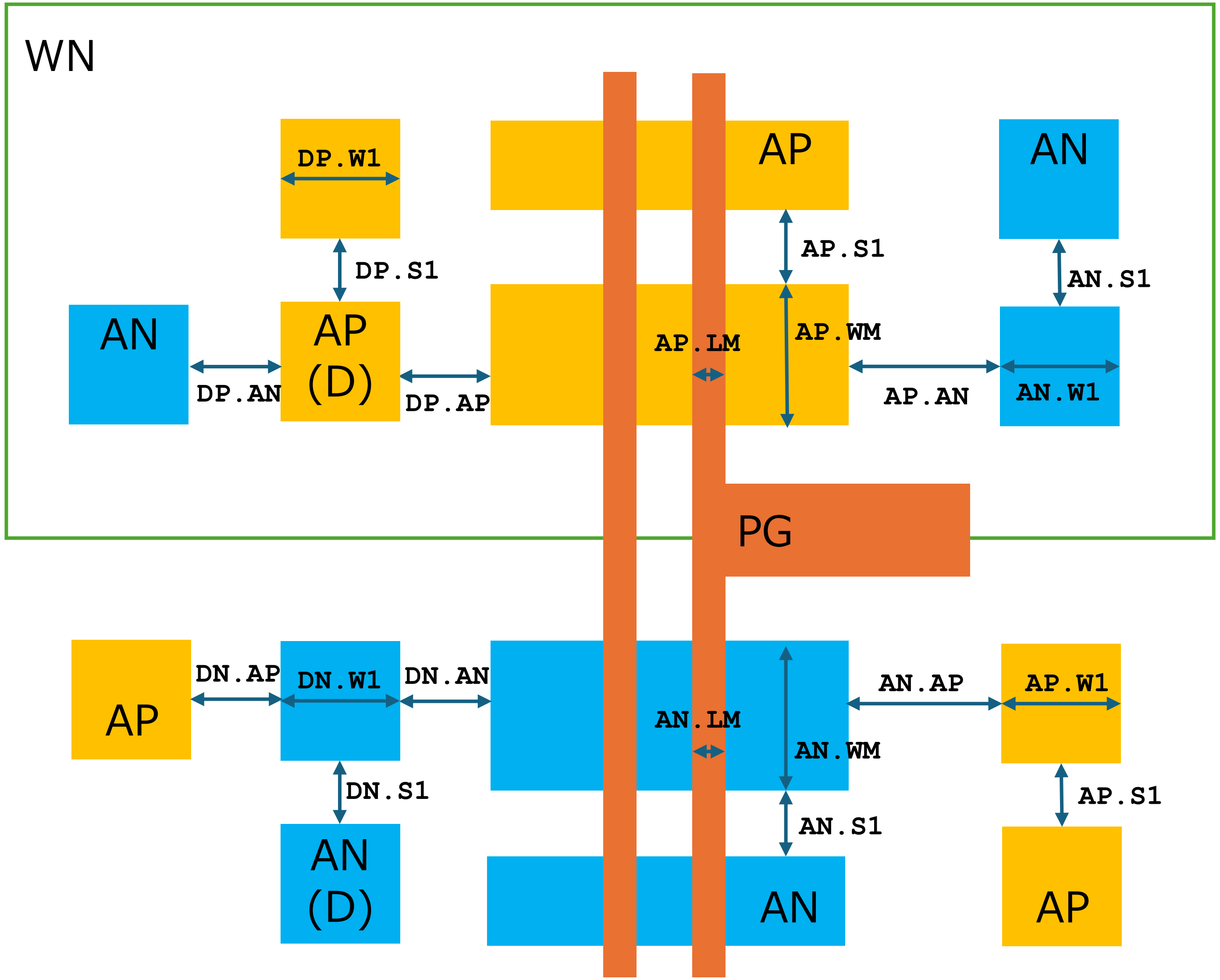
	Original MASK layers			Proposed Drawing layers			
	Name	NUM	TYPE	Name	NUM	TYPE	
1	PSUB	140	0	WN	140	0	
2	NW	36	0				MDP generatioed
3	HVNW	141	0				MDP generatioed
4	L	3	0				MDP generatioed
				AP	3	1	Active for P+
				AN	3	2	Active for N+
				AR	3	3	Active for resistor
				AC	3	4	Active for capacitance
5	NF	25	0				MDP generatioed
6	PF	26	0				MDP generatioed
7	CL	143	0				MDP generatioed
8	HPBE	144	0				No USE
9	HNBE	145	0				No USE
10	PBE	146	0				MDP generatioed
11	NBE	147	0				MDP generatioed
12	HPM	33	0				MDP generatioed
13	RHP	148	0				No USE
14	SG	8	0				
				PO	8	1	POLY transistor
				PR	8	2	POLY resistor
15	PM	35	0				MDP generatioed
16	NM	7	0				MDP generatioed
17	R	12	0				MDP generatioed
18	PSD	9	0				MDP generatioed
19	NSD	28	0				MDP generatioed
20	CONT	11	0	CO	11	0	
21	M1	13	0	M1	13	0	
22	TC	19	0	V1	19	0	
23	M2	20	0	M2	20	0	
24	PRO	14	0	PE	14	0	
	TXM1	48	0	TXM1	48	0	M1 Label
	PIN	48	1	PIN	48	1	M1 Pin
	TXM2	49	0	TXM2	49	0	M2 Label
	MASK	63	0	MASK	63	0	Waiver region
	MEAS	63	1	MEAS	63	1	Measurement region
	SCRIBE	80	0	SCRIBE	80	0	Waiver region (EDGE SEAL)
	prBoundary	235	0	prBoundary	235	0	Cell boundary

WN to AP/AN/AR/AC related



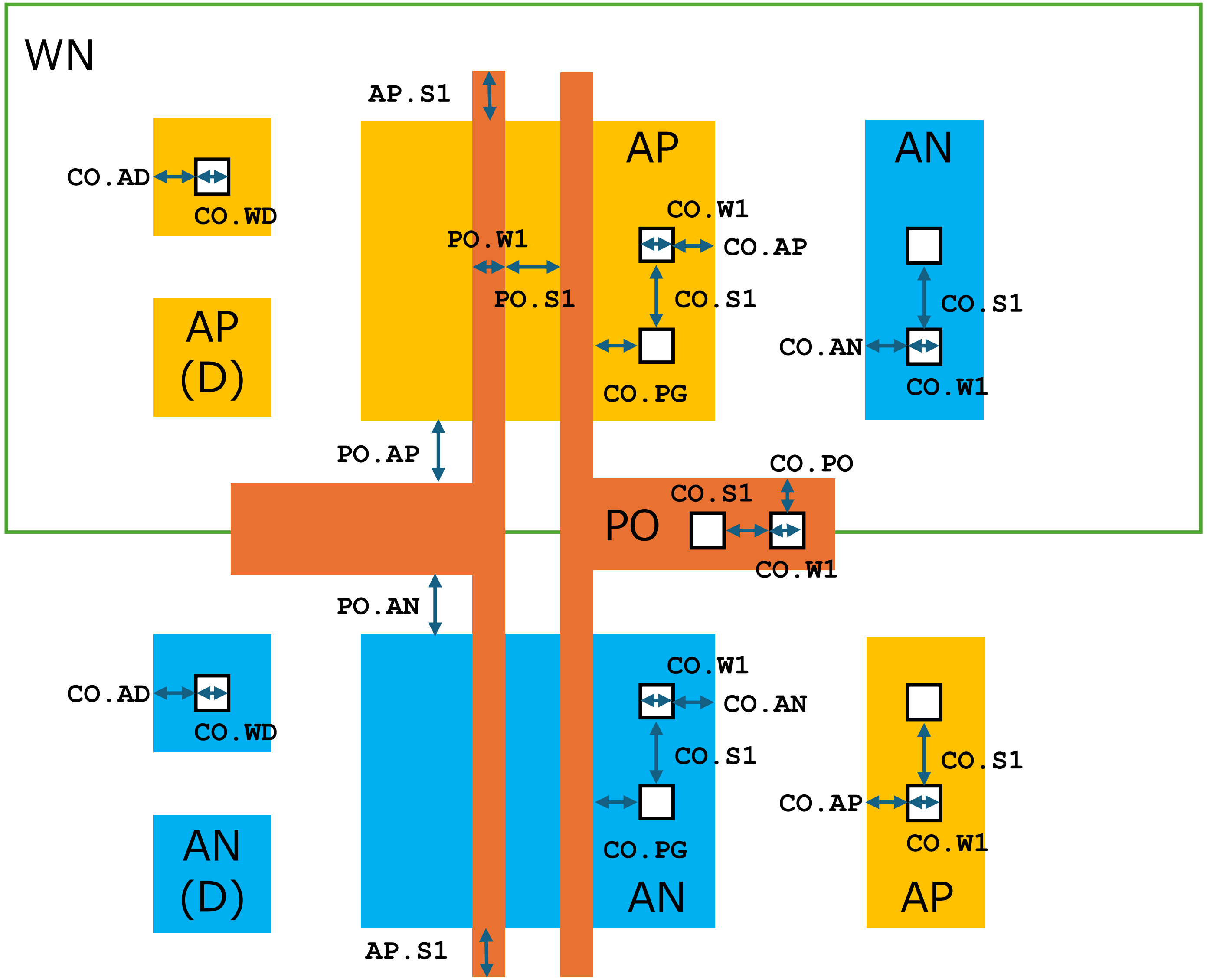
Name	Description	Rule	MAX
WN.W1	WN Wmin	8.0	inf
WN.S1	WN-WN Smin	12.0	inf
WN.S2	WN-WN(R) Smin	9.5	inf
WN.S3	WN-WN(C) Smin	12.0	inf
WN.S4	WN(R)-WN(C) Smin	9.5	inf
WN.S5	WN(R)-WN(R) Smin	8.0	inf
WN.S6	WN(C)-WN(C) Smin	12.0	inf
WN.AP	WN-AP Smin	10.0	inf
WN.AN	WN-AN Smin	5.0	inf
AR.W1	AR Wmin/max	2.8	20.0
AR.S1	AR Smin	4.0	inf
AR.XY	AR corner cut	1.0	inf
AR.WR	AR-WN(R) Emin	10.0	inf
AN.WR	AN-WN(R) Emin	5.0	inf
AR.AN	AR-AN Smin	4.0	inf
AC.W1	AC Wmin/max	28.5	120.0
AC.S1	AC Smin	???	???
AC.WC	AC-WN(C) Emin	???	???
AN.WC	AN-WN(C) Emin	3.0	inf
AC.AN	AC-AN Smin	2.8	inf
AC.R1	AN must surround AC		

MOS transistor related



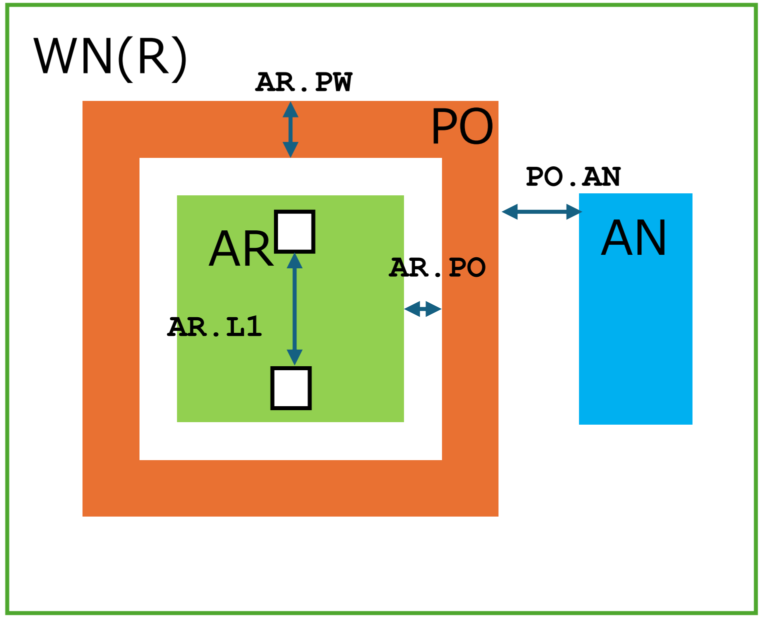
Name	Description	Rule	MAX
AP.W1	AP Wmin	1.4	inf
AP.S1	AP Smin	1.4	inf
AP.AN	AP-AN Smin	2.8	inf
AP.WM	PMOS Wmin/max	3.4	60.0
AP.LM	PMOS Lmin/max	1.0	30.0
AP.WN	AP-WN Emin	7.0	inf
DP.W1	AP Wmin	1.4	inf
DP.S1	DP Smin	???	???
DP.AP	DP-AP Smin	2.8	inf
DP.AN	DP-AN Smin	2.8	inf
AN.W1	AN Wmin	1.4	inf
AN.S1	AN Smin	1.4	inf
AN.AP	AN-AP Smin	2.8	inf
AN.WM	NMOS Wmin	3.4	60.0
AN.LM	NMOS Lmin/max	1.0	30.0
AN.WN	AN-WN Emin	5.0	inf
DN.S1	DN Smin	???	???
DN.AP	DP-AP Smin	2.8	inf
DN.AN	DP-AN Smin	2.8	inf

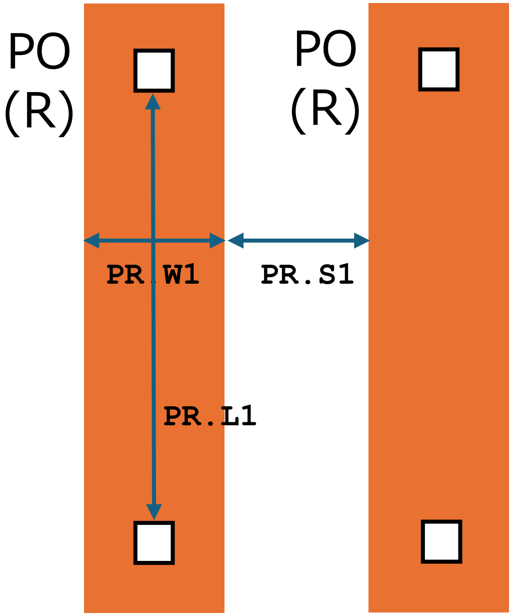
Contact and Poly related

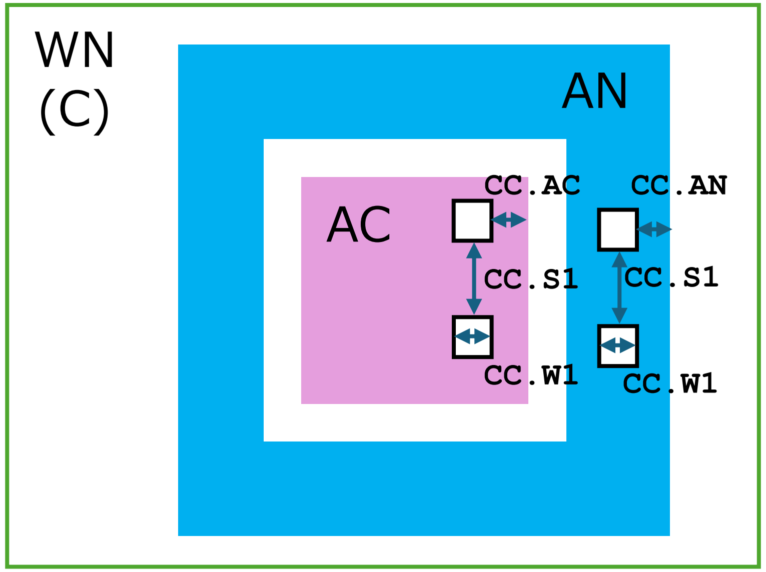


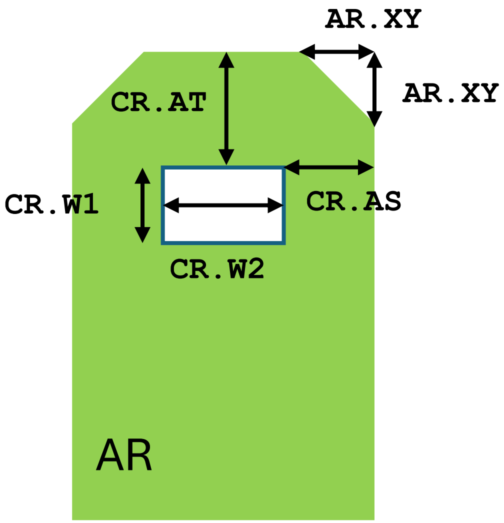
Name	Description	Rule	MAX
PO.W1	PO Wmin	1.0	30.0
PO.S1	PO Smin	1.2	inf
PO.AP	PO-AP Smin	0.4	inf
PO.AN	PO-AN Smin	0.4	inf
PO.EM	MOS Endcap min	1.2	inf
CO.W1	CO Wmin	1.0	inf
CO.S1	CO Smin	1.0	inf
CO.WD	CO(D) Width	1.2	inf
CO.AP	CO-AP Emin	0.8	inf
CO.AN	CO-AN Emin	0.8	inf
CO.PG	CO-PO(G) Smin	1.0	inf
CO.PO	CO-PO Emin	0.8	inf
CO.AD	CO-AD Emin	1.2	inf

Resistor(AR/AS) and Capacitor(CSIO) related



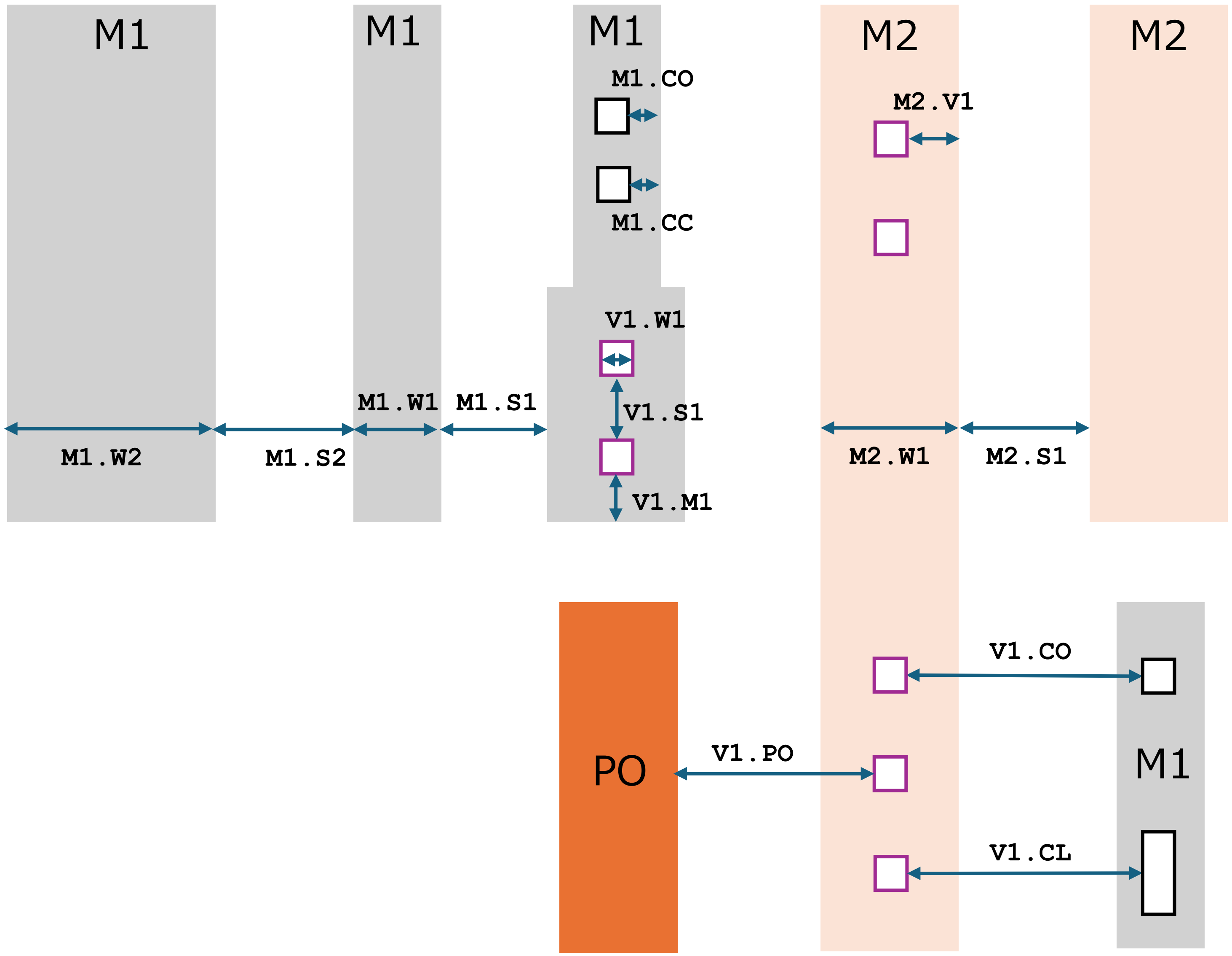






Name	Description	Rule	MAX
AR.L1	AR Lmin/max	13.0	100.0
AR.PO	PO-AR Smin	1.0	inf
AR.PW	PO(AR) Wmin	2.0	inf
PO.L1	PO(AR) Lmin/max	20.0	200.0
PO.AN	PO(AR)-AN Smin	???	???
PO.R1	PO must surround AR		
PO.R2	PO must tie-down		
PR.W1	PO(R) Wmin/max	4.0	20.0
PR.L1	PO(R) Lmin/max	20.0	100.0
PR.S1	PO(R) Smin	2.0	inf
CR.W1	CO(RR) Width	1.0	inf
CR.W2	CO(RR) Wmin	1.0	inf
CR.AT	CO(RR)-AR(Top) Emin	1.5	inf
CR.AS	CO(RR)-AR(Side) Emin	1.0	inf
CC.W1	CO(C) Wmin	1.2	inf
CC.S1	CO(C) Smin	1.2	inf
CC.AC	CO(C)-AC Emin	2.9	inf
CC.AN	CO(C)-AN Emim	1.2	inf

M1/V1/M2 related



Name	Description	Rule	MAX
M1.W1	M1 Wmin	1.8	inf
M1.S1	M1 Smin	1.4	inf
M1.CO	CO-M1 Fmin	0.8	inf
M1.CC	CO(C)-M1 Fmin	1.2	inf
M1.WW	M1(W) Wmin	10.0	inf
M1.SW	M1(W) Smin	2.0	inf
V1.W1	V1 Windth	1.4	inf
V1.S1	V1 Smin	1.5	inf
V1.M1	V1-M1 Emin	1.0	inf
V1.PO	V1-PO Smin	1.2	inf
V1.CO	V1-CO Smin	1.0	inf
V1.CL	V1-CO(L) Smin	1.4	inf
M2.W1	M2 Wmin	3.0	inf
M2.S1	M2 Smin	2.0	inf
M2.V1	V1-M2 Fmin	1.0	inf

質問・確認事項

2025/12/25

DIODE

リファレンス・マニュアルTable I-1-1によればダイオードはP/NどちらもPM+NMインプラと記載

DP：PM+NM
DN：PM+NM

リファレンス・マニュアル図6.1/6.2にも同様の指示あるが、

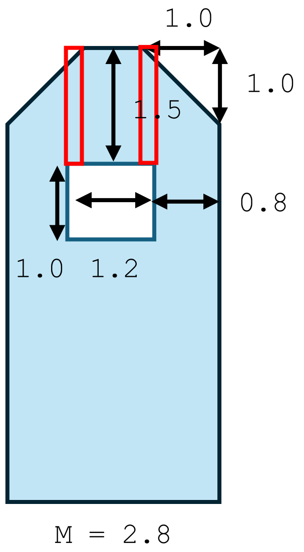
リファレンス・マニュアル図6.12/6.13に(ESD)では

DP：NM
DN：PM

と指示がある。本当はどちらでしょう？違う場合は、Spiceモデルを公開頂けますと幸いです。

また、ESD用NMOSのSPICEモデルも公開頂けますと幸いです。

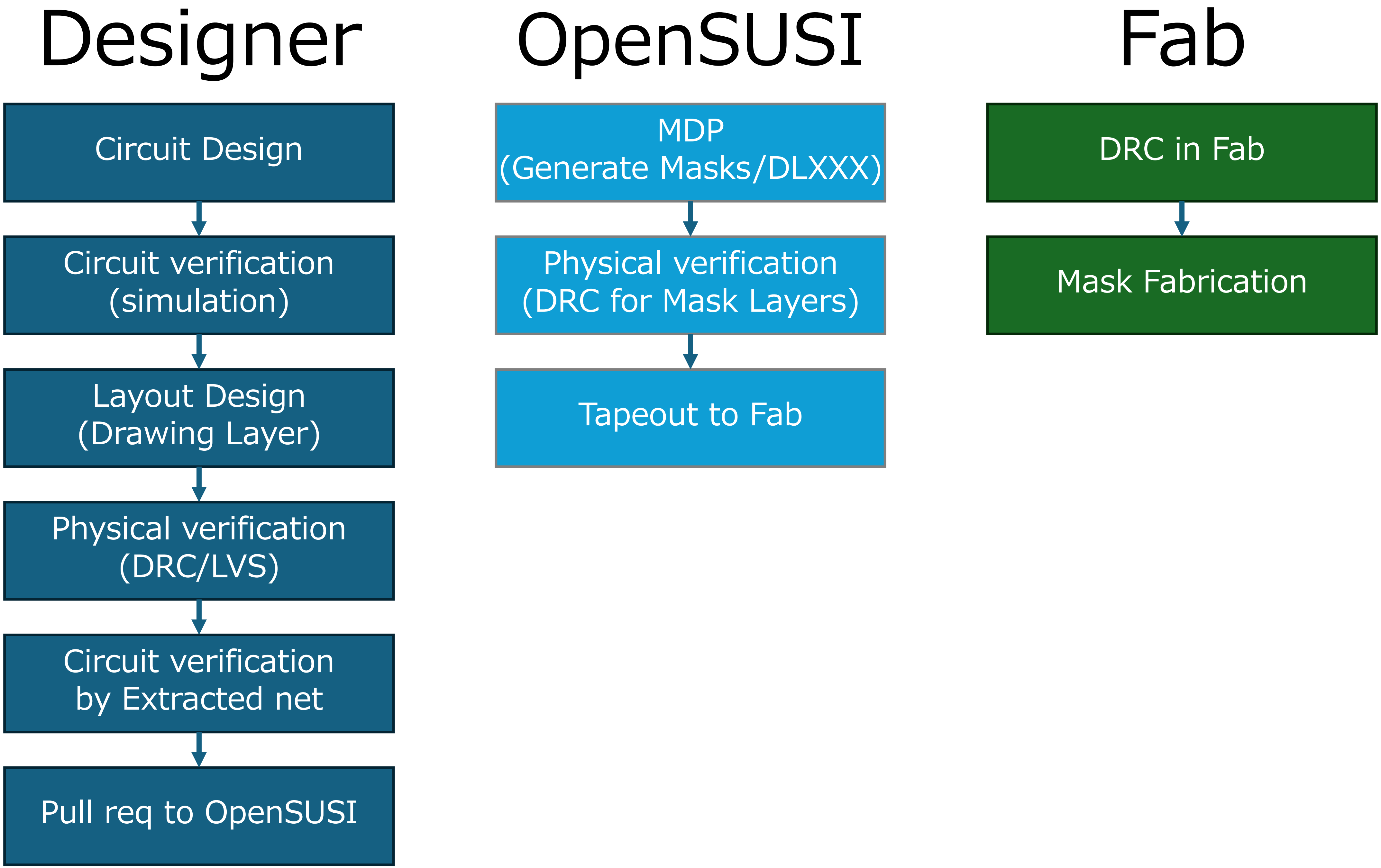
Design Rule for RR details



1. RR抵抗幅2.8umの場合：
   ERRR14: 0.8um
   RRのときは可変長コンタクト
   ERRR08： 1.5um
   赤枠のエリアで ERR08 がエラー

Design to Tapeout Process

Mask Data Preparation


Designer
Circuit Design
Circuit verification
(simulation)
Layout Design
(Drawing Layer)
Physical verification
(DRC/LVS)
Circuit verification
by Extracted net
Pull req to OpenSUSI
OpenSUSI
MDP
(Generate Masks/DLXXX)
Physical verification
(DRC for Mask Layers)
Tapeout to Fab
Fab
DRC in Fab
Mask Fabrication